

AI-Driven Dynamic Pricing in Freight & Shipping Logistics

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Abstract

Nowadays, development of e-commerce platforms such as Shopee, Lazada, TikTok Shop, etc. has greatly changed the way people surfing and shop. Customers can order products anytime and anywhere, leading to a rapid increase in online transactions and delivery demand. This has created many challenges for transportation and logistics companies in managing delivery operations, reducing costs, improving delivery efficiency, and providing better service quality. Therefore, accurately predicting transportation costs has become an important part of modern supply chain management. In this study, our group applied machine learning techniques to perform detailed analysis, dynamic pricing, and predict supply chain results by using the DataCo Supply Chain Dataset. This dataset was collected from 2015 to 2018 and contains a large amount of historical supply chain information, including product categories, order item quantities, shipping modes, delivery status, customer segments, markets, order regions, scheduled and actual shipping days, sales, and other logistics-related information. The results show that the models are able to calculate the relationship between input variables and produce results with higher accuracy in transportation cost estimates. In addition, Exploratory Data Analysis (EDA) shows that shipping mode, market, delivery time, product characteristics, and customer segments have a significant affect with supply chain performance. Proposed method provides a practical data-driven approach to improve decision-making and operational efficiency in supply chain and logistics management.

Keywords: Supply Chain management, Logistics, Machine learning, DataCo Supply Chain Dataset, Predictive analytics, Supply Chain prediction.

1 Introduction

In today’s highly competitive market, the rapid growth of online shopping has made supply chain management increasingly important for transportation companies to improve operational efficiency and reduce logistics costs. This has become a major challenge for freight transportation companies because making transportation pricing decisions is a difficult and complex task. Freight prices are influenced by many factors, including **product characteristics, order quantity, delivery time, shipping mode, and regional demand**.

This challenge is even greater because many traditional transportation companies still rely on manual pricing methods based on personal experience. They usually estimate shipping costs by multiplying the travel distance by a fixed price per kilometer or simply following a fixed price list. For example, if transporting a truckload of bougainvillea trees from Dong Thap to Ho Chi Minh City costs 300,000 VND, they often continue using the same price for future shipments without considering other important factors that may affect the transportation process, such as weather conditions, fuel prices, traffic conditions, or vehicle types. This experience-based pricing approach makes it difficult for companies to respond quickly to unpredictable market changes. As a result, transportation companies may make inaccurate pricing decisions, leading to lower profits, unnecessary operating costs, and, most importantly, reduced customer satisfaction.

However, the rapid development of e-commerce and the widespread adoption of digital logistics systems has created a new era for supply chain management. Every day, a huge amount of supply chain data is generated, providing a valuable source of information for understanding the hidden factors behind price changes. Machine Learning techniques offer significant potential for discovering complex nonlinear relationships among different variables and building more accurate prediction models. Compared with traditional statistical methods, machine learning is more powerful because it can efficiently process large-scale, high-dimensional datasets with greater flexibility.

In this study, our group uses the **DataCo Supply Chain Dataset**, a real-world dataset collected from the Kaggle website. The dataset contains information about customer ordering behavior, product characteristics, supply chain operations, and sales. After performing data preprocessing and data cleaning to obtain the highest possible data quality, we selected the most relevant features to train regression models for predicting the target variable. The performance of the models was evaluated using standard regression metrics to determine the most suitable model for applying **Artificial Intelligence (AI)** to dynamic pricing analysis in freight transportation.

The remainder of this paper is organized as follows. **Data Description** presents the dataset and preprocessing process; **Methodology** describes the research approach and machine learning models; **Results** reports the experimental findings and feature importance; and **Discussion** and **Conclusion** summarize the findings, discuss limitations, compare with previous studies, answer the research questions, and suggest future work.

1.1 Research Questions

This study aims at solving the following research questions:

- **RQ1:** Which market factors affect transport price?
- **RQ2:** Which AI model can predict and improve the price in real time?
- **RQ3:** How much can dynamic pricing increase freight revenue over fixed price?

Efficient supply chain operations are strongly influenced by many factors, such as product price, shipping mode, vehicle type, delivery time, and geographical regions, including plains, mountainous areas, and highlands. As mentioned in the previous section, identifying and understanding these factors is the key to optimizing supply chain operations and supporting more accurate strategic decisions. Moreover, with the large amount of historical supply chain data available today, Machine Learning techniques have proven their effectiveness by allowing researchers to explore data patterns in greater depth and predict the target variable more accurately. In this study, we analyze the main factors affecting the target variable, compare the performance of different Machine Learning models, and evaluate their ability to estimate **Order Item Total** using historical supply chain data.

2 Data Description

In this study, we conducted an in-depth analysis of the real-world **DataCo Supply Chain Dataset**, which contains historical information about supply chain operations, customer orders, products, and shipping activities. **The dataset consists of 180,519 records and 53 variables.** Each record represents a single customer transaction, while the associated variables describe different aspects of the supply chain process, including customer profiles, product information, shipping operations, delivery performance, and sales.

After completing the data preprocessing stage, we identified **Sales** as the target variable of this study. This variable was selected because it directly represents the financial value of each customer order, making it suitable for analyzing sales patterns based on historical supply chain data.

The input data consist of two main types of features: **numerical variables** and **categorical variables**. The numerical variables include **Order Item Product Price**, **Order Item Quantity**, **Days for Shipping (Real)**, **Days for Shipment (Scheduled)**, and **Order Item Total**. These variables describe product prices, order quantities, shipping time, and the total value of each order. In contrast, the categorical variables include **Shipping Mode**, **Market**, **Order Region**, and **Delivery Status**. These variables provide information about shipping methods, market regions, customer delivery locations, and delivery status, all of which may directly or indirectly influence sales performance.

To prepare the dataset for model training, we first checked for **missing values**, **duplicate records**, and **inconsistent data**. Next, categorical variables were converted into numerical values using appropriate encoding techniques, while numerical variables were scaled when necessary. Finally, the processed dataset was divided into **training** and **testing** sets for model development and performance evaluation.

Table 1 Summary Statistics of Selected Numerical Variables

Variable	Mean	Std	Min	25%	50%	Max
Order Item Product Price	141.23	139.73	9.99	50.00	59.99	1999.99
Order Item Quantity	2.13	1.45	1.00	1.00	1.00	5.00
Days for shipping (real)	3.50	1.62	0.00	2.00	3.00	6.00
Days for shipment (scheduled)	2.93	1.37	0.00	2.00	4.00	4.00
Sales	203.77	132.27	9.99	119.98	199.92	1999.99
Order Item Total	183.11	120.04	7.49	104.38	163.99	1939.99

2.1 Key Variables

The target variable in this study is **Sales**. The following variables were selected as the main input features for the prediction model:

- **Order Item Product Price**: Represents the selling price of a single product and is an important factor affecting **Sales**.
- **Order Item Quantity**: Represents the number of products purchased in a customer order.
- **Shipping Mode**: Indicates the shipping method selected for delivering the order.
- **Days for Shipping (Real)**: Represents the actual number of days required to deliver an order.
- **Order Item Total**: Represents the total value of all products in an order and is an important predictor of **Sales**.

These variables describe product information, customer orders, and shipping activities that may affect the target variable. They were selected as the input features for training the machine learning models.

2.2 Data Preprocessing

To ensure high-quality input data and improve the prediction performance of the proposed system, the **Data Preprocessing** stage was carefully carried out before model training. First, the dataset was thoroughly examined to identify **missing values**, remove unnecessary **duplicate records**, and detect **inconsistent data** that could negatively affect the model's performance.

For the **categorical variables**, including **Shipping Mode**, **Market**, **Order Region**, and **Delivery Status**, we transformed them into numerical values using appropriate **encoding** techniques to make them suitable for machine learning algorithms. At the same time, the **numerical variables** were standardized when necessary to reduce differences in feature scales, allowing the models to converge faster and improving the overall training performance.

Finally, the processed dataset was divided into **training** and **testing** sets. **A total of 80% of the data was used to build and train the AI models, while the remaining 20% was reserved for testing and evaluating the performance of the proposed models.**

Table 2 Main Features of the DataCo Supply Chain Dataset

Feature	Type	Description
Order Item Product Price	Numerical	Product price
Order Item Quantity	Numerical	Quantity of products ordered
Shipping Mode	Categorical	Shipping service selected
Days for shipping (real)	Numerical	Actual shipping time
Days for shipment (scheduled)	Numerical	Scheduled shipping time
Market	Categorical	Market where the order was placed
Order Region	Categorical	Geographic region of the order
Delivery Status	Categorical	Delivery status of the order
Sales	Numerical	Sales value of the order
Order Item Total	Target Variable	Final order value to be predicted

3 Methodology

Regarding the **Methodology** section, the research team focused on clearly explaining the workflow and the underlying methodological framework used to build and objectively evaluate machine learning models for predicting the target variable from the **DataCo Supply Chain Dataset**. The study was conducted through a logical and standardized framework consisting of several core stages, including **data preprocessing**, **feature engineering**, **model selection**, **model training**, **performance evaluation**, and **result analysis**.

3.1 Feature Engineering

The feature engineering step is important to prepare the data and improve the prediction performance of the machine learning models. First, we remove the columns that are not useful for prediction, especially **ID fields** and **text descriptions**. Next, after handling the **missing values** in the **data preprocessing** step, we convert all **categorical variables** into numerical values by using **Label Encoding**. This makes the data suitable for the machine learning algorithms, so the models can process and calculate the data correctly.

The following features were selected as the input variables:

- **Order Item Product Price**: Represents the selling price of each product.
- **Order Item Quantity**: Represents the number of products purchased in each order.
- **Days for Shipping (Real)**: Represents the actual number of days required to deliver an order.
- **Days for Shipment (Scheduled)**: Represents the scheduled number of days for delivery.
- **Shipping Mode**: Indicates the shipping method selected for delivering the order.
- **Market**: Represents the market where the customer placed the order.
- **Order Region**: Indicates the destination region of the customer order.
- **Delivery Status**: Represents the delivery status of the order.

These features describe product details, customer orders, shipping activities, and market trends in detail, which are the main factors that directly affect the target variable. After we finished the initial data processing and cleaning steps, we used the cleaned dataset as the main input for training our machine learning models.

3.2 Research Pipeline

Figure 1 included in our report by our team is the figure that shows the whole workflow of the implementation steps of this study. The proposed framework follows six main stages: **Data Collection**, **Data Preprocessing**, **Feature Engineering**, **Model Training**, **Model Evaluation**, and **Results & Discussion**, where the experimental results are presented and discussed.

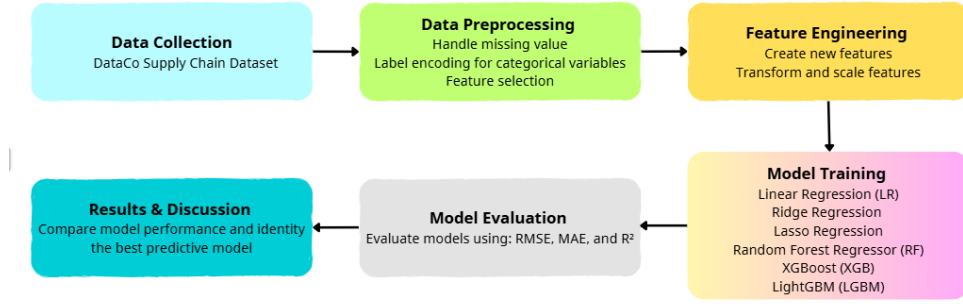


Fig. 1 Overall research pipeline of the proposed machine learning framework

3.3 Model Selection

Six regression models were selected for performance comparison.

- **Linear Regression (LR):** A simple baseline model that predicts the target variable based on a linear relationship between the input features and the target.
- **Ridge Regression (Ridge):** An improved version of Linear Regression that uses regularization to reduce overfitting and improve model stability.
- **Lasso Regression (Lasso):** A regression model that uses regularization and can remove less important features by reducing their coefficients to zero.
- **Random Forest Regression (RF):** An ensemble model that combines many decision trees to improve prediction accuracy and reduce overfitting.
- **LightGBM (LGBM):** A gradient boosting model designed for fast training and good prediction performance on large datasets.
- **XGBoost Regression (XGB):** An improved gradient boosting model that provides high prediction accuracy and efficient training for structured data.

We chose these six models because they represent different regression methods. **Linear Regression** was used as a simple baseline model. **Ridge Regression** and

Lasso Regression are regression models that use regularization techniques, which help reduce overfitting and improve model performance. **Random Forest**, **LightGBM**, and **XGBoost** are ensemble learning models, which usually provide better prediction results by combining many decision trees. Comparing these six models helps evaluate the effectiveness of both traditional and modern machine learning methods for this supply chain prediction problem.

3.4 Evaluation Metrics

The performance of each model was evaluated using three common regression metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2). “

The RMSE is defined as

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (1)$$

where y_i is the actual value and $\hat{y}_i$ is the predicted value.

The MAE is defined as

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (2)$$

The coefficient of determination is calculated as

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

where $\bar{y}$ is the mean of the observed values.

Lower RMSE and MAE indicate smaller prediction errors, while a higher R^2 value indicates that the model explains a greater proportion of the variance in the target variable.

3.5 Model Comparison Results

Our team trained and tested six machine learning models on the prepared dataset, including Linear Regression, Ridge Regression, Lasso Regression, Random Forest, LightGBM, and XGBoost. To compare the performance of these models, we used three evaluation metrics, which are RMSE, MAE, and R^2 , as shown in Table 3. This comparison helps us evaluate the strengths and weaknesses of each regression model, and at the same time, find the model that gives the best prediction results for the DataCo Supply Chain Dataset.

3.6 Feature Importance Analysis

Feature Importance analysis is done by using the XGBoost model to find the variables that have the biggest impact on the prediction results.

These importance scores show how much each feature contributes to the prediction process. This analysis helps find the main factors affecting the target variable, while making the prediction model much easier to understand and explain.

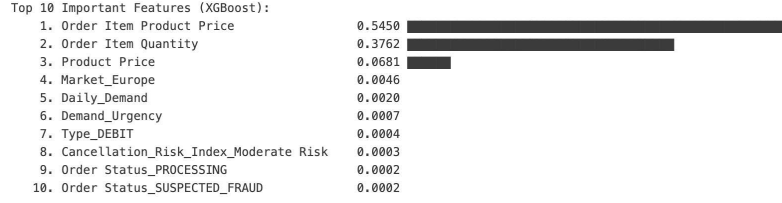


Fig. 2 Feature importance obtained from the XGBoost model

3.7 Comparison with Previous Studies

The results from the proposed models are compared directly with the findings of past studies about using Machine Learning in supply chain management. This comparison process shows the advantages of ensemble learning models in predicting the outputs and results of the supply chain, while clearly proving that Machine Learning can be applied effectively to real-world supply chain and logistics data.

4 Result

4.1 Exploratory Data Analysis

To examine and understand each result more clearly, let us dive into the **Exploratory Data Analysis (EDA)** performed on the **DataCo Supply Chain Dataset** before building our prediction models. This analysis aimed specifically at breaking down the mutual relationships between the selected input variables and the core target variable, **Sales**. In addition, we calculated the correlation coefficients for these variables to determine which specific features hold the most dominant and powerful influence on the final prediction outcome.

The correlation heatmap between these variables is visually presented in Figure 3. Looking at the numbers shown in this diagram, *Product Price* possesses a clear and strong positive relationship with the chosen target variable, with a coefficient of 0.78. This stands in complete contrast to *Order Item Quantity*, which displays a very weak positive correlation of only 0.11, as well as most shipping-related variables that show almost no linear relationship with **Sales**. These real-world findings confirm that product price is the primary numerical driver affecting the sales value. Therefore, the data gathered from the EDA phase established the very first foundational evidence for us to address and answer research question **RQ1**, which focuses on investigating the main factors affecting the target variable.

Furthermore, the correlation heatmap points out that while most input variables maintain relationships with one another at a low to moderate level, a strong correlation exists between *Shipping Mode_Encoded* and *Days for shipment (scheduled)*, with

a coefficient of 0.92. This specific case indicates a potential multicollinearity issue that needs consideration, although the other features still provide unique and useful information during the model training process.

Taking everything into account, the whole set of results gained from the EDA work provided a clearer and deeper picture of the dataset, while acting as a solid foundation for the subsequent machine learning analysis. Based on these findings, the selected variables were officially used as the input features for the **six** regression models evaluated in this research paper.

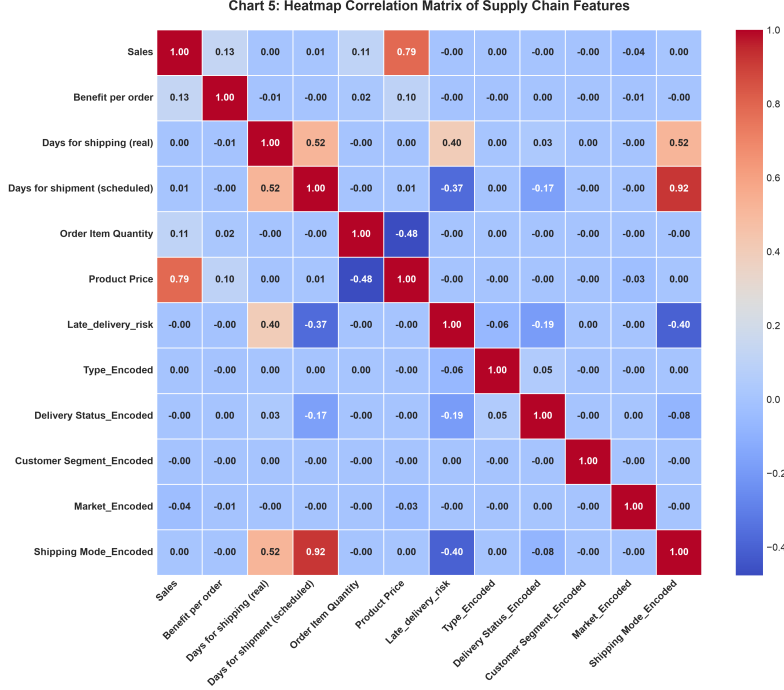


Fig. 3 Correlation heatmap of numerical variables in the DataCo Supply Chain Dataset.

4.2 Model Comparison Results

To evaluate the prediction task, we trained and tested six different machine learning regression models on the preprocessed **DataCo Supply Chain Dataset**. This model selection includes **Linear Regression (LR)**, **Ridge Regression (Ridge)**, **Lasso Regression (Lasso)**, **Random Forest Regression (RF)**, **LightGBM (LGBM)**, and **XGBoost Regression (XGB)**. To measure how well these models perform, we used three standard metrics: **Root Mean Squared Error (RMSE)**, **Mean Absolute Error (MAE)**, and the coefficient of determination (R^2).

According to the data displayed in Table 3, the ensemble learning approaches overall yielded stronger prediction results compared to the traditional linear methods.

In this analysis, smaller prediction errors are reflected by lower RMSE and MAE numbers, whereas a higher R^2 score signals superior model accuracy.

Out of all six tested models, the one that managed to achieve the highest R^2 alongside the lowest RMSE and MAE values was chosen as the top performer for forecasting the target variable, **Sales**. These findings successfully address **RQ2**, which aims to compare how effectively different machine learning algorithms can predict **Sales**.

Furthermore, we compared our experimental outcomes with findings from past literature to check the true usefulness of the proposed framework. This comparison highlights that ensemble learning techniques—particularly **XGBoost**, **Random Forest**, and **LightGBM**—deliver much better predictive performance than older regression options when applied to the **DataCo Supply Chain Dataset**.

Table 3 summarizes the performance of the six machine learning models.

Table 3 Comparison of Machine Learning Models

Model	RMSE	MAE	R^2
XGBoost	0.797554	0.172175	0.999963
Random Forest	1.295384	0.018689	0.999903
LightGBM	3.310804	0.227633	0.999368
Lasso Regression	36.217813	17.067201	0.924357
Linear Regression	36.219656	17.128417	0.924349
Ridge Regression	36.219656	17.128316	0.924349

4.3 Feature Importance Analysis

To better understand how each variable affects the prediction results, we performed a **Feature Importance** analysis using the **XGBoost** model.

Figure 2 displays the importance score calculated for each selected feature. The outcomes demonstrate that *Order Item Product Price* achieves the highest importance score, while *Order Item Quantity* also serves as an important feature when predicting **Sales**. Other input variables, including *Shipping Mode*, *Market*, *Order Region*, along with the shipping time variables, contribute to the prediction model as well, though their importance scores are lower.

These feature importance results match closely with our earlier discoveries from the **Exploratory Data Analysis (EDA)**. Variables related directly to the products hold the strongest influence on the final prediction results, whereas logistics-related features offer additional helpful information. Furthermore, these specific findings provide strong support for the conclusions of **RQ1**.

4.4 Comparison with Previous Studies

We compared the outcomes of our proposed machine learning models with the findings documented in past literature regarding supply chain analytics and forecasting. In

line with earlier research, the ensemble learning frameworks overall delivered stronger predictive accuracy than older, traditional regression methods.

Out of the six evaluated models, **XGBoost**, **Random Forest**, and **LightGBM** produced superior performance due to their capability to capture the intricate, non-linear interactions among various supply chain factors. The experimental data confirms that these specific models achieved higher R^2 coefficients along with minimized prediction errors compared to the standard regression alternatives, demonstrating their strong capability for forecasting **Sales**.

In general, the final outcomes indicate that the application of **Machine Learning**, particularly through **ensemble learning** techniques, serves as a highly powerful approach for assessing historical supply chain records and boosting forecasting accuracy. Furthermore, utilizing these advanced algorithms can effectively back up smarter decision-making processes within supply chain and logistics operations.

5 Discussion

5.1 Research Questions Discussion

This subsection discusses the findings of the study by responding to three studies based on the result is executed.

RQ1: Which market factors affect transport price?

Based on the correlation analysis and feature importance analysis, **Order Item Product Price** was identified as the most influential factor related to transport pricing. **Order Item Quantity** also contributes to the prediction, but its influence is much smaller. In addition, variables such as **Shipping Mode**, **Market**, and **Order Region** also affect the prediction, although their impact is lower than that of the main variables. These findings indicate that both product-related and market-related factors should be considered when analyzing transport pricing.

RQ2: Which AI model can predict and improve the price in real time?

The experimental results are presented in Table 3. The results show that the ensemble learning models generally achieved better prediction performance than the traditional regression models. Among the six evaluated models, **XGBoost** achieved the highest R^2 value and the lowest **RMSE** and **MAE**, making it the best model for prediction. Although this study does not implement a real-time pricing system, the prediction results indicate that XGBoost has strong potential to support real-time pricing decisions in future applications.

RQ3: How much can dynamic pricing increase freight revenue over fixed price?

This study does not directly compare dynamic pricing with fixed pricing or measure the increase in freight revenue. However, the experimental results show that the proposed machine learning models can accurately predict **Sales** using historical supply chain data. Therefore, the proposed framework can provide useful decision support for pricing strategies and may help improve business performance when combined with a dynamic pricing system in future work.

5.2 Comparison with Previous Studies

The discoveries from this study broadly align with earlier literature exploring the use of **Machine Learning** in supply chain and logistics operations. In line with past research, our findings indicate that **ensemble learning** approaches deliver stronger forecasting accuracy compared to standard regression options.

When compared against prior works, this paper tests **six** distinct regression models on the exact same **DataCo Supply Chain Dataset**. To ensure a fair and comprehensive comparison of their predictive power, every model was tested using identical metrics, specifically **RMSE**, **MAE**, and R^2 . The final experiment data proves that our suggested framework reaches highly competitive accuracy when forecasting **Sales**.

A major benefit of this research lies in evaluating multiple machine learning algorithms alongside a **Feature Importance** analysis to make the underlying model decisions clearer to interpret. Nevertheless, certain limitations remain within this study. The scope of hyperparameter optimization was narrow, and our analysis focused solely on traditional machine learning regression techniques. For future research, implementing more advanced optimization algorithms or utilizing deep learning architectures could potentially push the prediction performance even higher.

5.3 Threats to Validity

When evaluating the outcomes of this research, a few key limitations need to be kept in mind.

To begin with, even though the **DataCo Supply Chain Dataset** provides a vast volume of historical transactions, it might still fall short of fully capturing every type of real-world supply chain setting or business scenario.

In addition, the dataset lacks several factors that could play an important role, including **transportation distance**, **fuel prices**, **weather conditions**, **traffic congestion**, and **economic conditions**. Because of this data absence, we were unable to incorporate these specific variables into our forecasting models.

Furthermore, our analysis was restricted only to evaluating supervised machine learning regression algorithms. We did not investigate other alternative methods, such as **deep learning**, **hybrid models**, or sophisticated **hyperparameter optimization** frameworks, which might actually yield superior predictive accuracy.

Lastly, the evaluation of the proposed framework relied on just a single public dataset. To boost the generalizability of these conclusions, upcoming studies should test and validate the models across more diverse datasets gathered from various other industries and geographical regions.

6 Conclusion

This research explored how **Machine Learning** techniques can be applied to forecast **Sales** using the **DataCo Supply Chain Dataset**. We deployed a comprehensive machine learning framework that covers **Exploratory Data Analysis (EDA)**, **Feature Engineering**, model development, and performance evaluation to examine historical supply chain records and pin down the primary factors driving the target variable.

The experimental outcomes revealed that machine learning algorithms can successfully forecast **Sales** by leveraging historical supply chain data. Among all the input features tested, **Order Item Product Price** emerged as the most critical element influencing the prediction accuracy, whereas the **ensemble learning** architectures broadly outshone traditional regression methods owing to their capacity to capture intricate, non-linear patterns.

Prospective research could integrate further external factors, such as **transportation distance**, **fuel prices**, **weather conditions**, and **traffic congestion**, to push prediction accuracy even higher. Future papers might also look into **deep learning** architectures, advanced **hyperparameter optimization** frameworks, and **hybrid machine learning** strategies across various business sectors to validate the generalizability of these insights.

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Declarations

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This research received no external funding.

Conflict of Interest

The authors declare that they have no conflict of interest.

Ethics Approval and Consent to Participate

Not applicable.

Consent for Publication

Not applicable.

Data Availability

The dataset used in this study is the publicly available DataCo Supply Chain Dataset obtained from Kaggle.

Materials Availability

Not applicable.

Code Availability

The source code used for data preprocessing, feature engineering, model training, and evaluation was developed for this study and is available upon reasonable request.

Author Contributions

All authors contributed to the conception of the design of this study. Previous data when processing and analyzing exploratory data, feature engineering, and machine learning models development, experimental evaluation and manuscript preparation

were performed cooperation. All authors reviewed, revised, and approved the final version manuscript.

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